

PORTFOLIO SAMPLE — This document was independently authored for demonstration purposes only, based on GE Vernova’s public information about g³ (g-cubed) SF6-free gas technology. It is not an official GE Vernova document and is not affiliated with, endorsed by, or sourced from GE Vernova. It does not cover any specific breaker model’s real ratings, intervals, or OEM procedures. Always follow the OEM maintenance manual, gas safety data sheet (SDS), and site protection/isolation procedures for the specific breaker being serviced.

GE VERNOVA — GRID SOLUTIONS (PORTFOLIO SAMPLE)
PREVENTIVE MAINTENANCE WORK INSTRUCTION
g³ SF6-Free Circuit Breakers
(g-cubed gas insulation technology)
WI-CB-G3-PM-2212 | Revision 01 | Effective Date: 2026-07-31

Document Control

Document No.	WI-CB-G3-PM-2212
Title	Preventive Maintenance Work Instruction — g ³ SF6-Free Circuit Breakers
Equipment Class	HV dead tank and live tank circuit breakers using g ³ (CO2 / O2 / fluoronitrile) insulating and switching gas
Revision	01
Effective Date	2026-07-31
Owner	Site Maintenance — Substation / Switchgear

1. Purpose and Scope

This work instruction defines the standard procedure for performing routine and periodic preventive maintenance on g³ (“g-cubed”) SF6-free high-voltage circuit breakers, so that dielectric integrity, mechanical reliability, and control circuit health are confirmed and maintained between major overhauls.

This instruction covers routine annual preventive maintenance and periodic diagnostic testing (timing and contact resistance). It does not cover internal tank inspection, contact replacement, or major overhaul procedures, which follow the OEM’s model-specific major maintenance manual and are outside this instruction’s scope.

NOTE: For diagnosing an active fault or alarm condition rather than performing scheduled maintenance, refer to the companion document TS-CB-G3-2211, g³ (g-cubed) SF6-Free Circuit Breaker Troubleshooting Manual.

2. Maintenance Interval Overview

Actual intervals depend on the specific breaker model, duty cycle, and number of operations — confirm the OEM-specified schedule for the unit being serviced before applying the intervals below as a default.

Task Category	Typical Interval	Covered In
Visual and gas system inspection	Annually, or per condition-monitoring alarm	Section 7 — Routine Preventive Maintenance
Mechanism function and control circuit test	1–3 years, per OEM schedule and duty	Section 7 — Routine Preventive Maintenance

Task Category	Typical Interval	Covered In
	cycle	
Timing and contact resistance test	3–6 years, or after a specified number of operations	Section 8 — Periodic Diagnostic Testing
Internal inspection / major overhaul	Per OEM interval or operation-count threshold (often 10+ years or a set number of fault interruptions)	OEM major overhaul procedure — outside the scope of this instruction

3. Required Personal Protective Equipment (PPE)

- Safety glasses with side shields; face shield for any work involving compressed gas fittings
- Insulated/dielectric gloves rated for the voltage class, where control circuit work requires it
- Flame-resistant (FR) clothing per the site's arc-flash policy
- Steel-toed safety footwear
- Hearing protection within posted high-noise zones
- Respiratory protection per the g³ SDS if opening any compartment that may contain arc decomposition byproducts

4. Required Tools and Test Equipment

- g³-compatible gas density/pressure gauge, confirmed rated for g³'s mixed-gas composition
- Insulation resistance (megohmmeter) tester
- Micro-ohmmeter for contact resistance testing
- Circuit breaker timing analyzer
- Trip/close coil current signature test set
- Torque wrench set matching the breaker's structural and electrical connection hardware
- Calibrated multimeter
- OEM instruction manual and maintenance record/history for the specific breaker
- Lockout/tagout kit and grounding equipment per site procedure

5. Safety Precautions

WARNING: g³ breakers store significant mechanical energy (charged springs, hydraulic/pneumatic accumulators) even when electrically isolated. Never work inside a mechanism compartment without first discharging or blocking stored energy per the OEM manual.

- Do not begin work until a valid switching order/permit has been issued, the breaker is confirmed open, isolated, and grounded per site procedure, and lockout/tagout is applied.
- Treat all control and trip circuits as energized until verified de-energized with a calibrated meter.
- Consult the g³ gas Safety Data Sheet (SDS) before opening any gas compartment, particularly one that has experienced arcing — do not assume SF₆ handling guidance applies without checking, since byproduct chemistry differs.

- Use appropriate arc-flash PPE per the site's arc-flash study/label for any work near energized conductors.
- Never bypass a protection or interlock function without following the site's management of change (MOC) and authorized bypass process.

6. Pre-Maintenance Checks

1. **Review maintenance history.** Check the breaker's maintenance record for open items, prior test results, and any trending gas density or timing data.
2. **Obtain permit and isolate.** Obtain the switching order/work permit, isolate and ground the breaker, and apply lockout/tagout.
3. **Notify control room.** Inform the control room that the breaker is being taken out of service for scheduled maintenance.
4. **Record baseline conditions.** Record ambient temperature, current gas density reading, and breaker position before beginning work.

7. Routine Preventive Maintenance (Annual)

5. **Visual inspection — exterior.** Inspect bushings, tank, and support structure for cracking, corrosion, contamination, or physical damage.
6. **Visual inspection — gas system.** Check gas density gauge reading against the OEM temperature-compensated curve; inspect fittings and valves for signs of leakage (frost, staining, audible leak).
7. **Check electrical connections.** Inspect and torque-check accessible electrical connections and ground connections to OEM specification.
8. **Inspect the operating mechanism cabinet.** Check for moisture ingress, corrosion, loose wiring, and heater/thermostat function (if fitted) to prevent condensation damage.
9. **Function-test the mechanism.** Perform a local open/close operation (per permit and OEM procedure) and confirm smooth operation, correct indication, and normal spring/hydraulic/pneumatic recharge behavior.
10. **Test trip and close coils.** Measure trip and close coil resistance/current signature and compare to OEM baseline values.
11. **Verify control circuit supervision.** Confirm any trip/close circuit supervision relay or alarm is functioning and not indicating a fault.
12. **Check auxiliary switch alignment.** Verify breaker position indication (52a/52b contacts) agrees with actual mechanical position.
13. **Clean as needed.** Clean accessible exterior surfaces, cabinet interiors, and cooling/ventilation openings per OEM guidance.
14. **Record results.** Record all readings, test results, and observations in the maintenance record, noting any values trending toward OEM limits.

8. Periodic Diagnostic Testing

Perform the following at the OEM-specified interval (typically several years, or after a defined number of operations), in addition to the routine annual tasks in Section 7.

- 15. **Contact resistance test.** Measure main contact resistance using a micro-ohmmeter and compare to OEM baseline and acceptance criteria.
- 16. **Timing test.** Perform an open/close timing test using a breaker timing analyzer; compare contact travel and timing curves to OEM specification.
- 17. **Insulation resistance test.** Measure insulation resistance across appropriate points per OEM procedure, accounting for the breaker's open/closed state as specified.
- 18. **Gas moisture check.** If supported by installed instrumentation or a portable analyzer, check gas moisture/dew point against OEM acceptance limits.
- 19. **Compare against trend data.** Compare current results to the breaker's historical trend rather than evaluating a single reading in isolation — a gradual trend toward a limit is often more informative than one data point.
- 20. **Escalate out-of-tolerance results.** If any result falls outside OEM acceptance criteria, do not return the breaker to service until the finding has been reviewed by engineering.

9. Post-Maintenance Verification

- ☐ All access panels and compartments are closed, sealed, and secured
- ☐ Gas density reads within the OEM-specified range for current ambient temperature
- ☐ No visible leaks, damage, or contamination remain unresolved
- ☐ Mechanism operates smoothly through a full open/close cycle with correct indication
- ☐ Trip and close coil test results are within OEM acceptance criteria
- ☐ All electrical connections are torqued to specification
- ☐ No tools, materials, or hardware remain inside any compartment
- ☐ All test results and observations are recorded in the maintenance record
- ☐ LOTO removed only after the breaker is confirmed fully reassembled and verified
- ☐ Control room notified prior to returning the breaker to service
- ☐ Work permit closed

10. Revision History

Rev.	Date	Author	Description of Change
01	2026-07-31	M. Langley, Tech. Writer	Initial release covering routine annual PM and periodic diagnostic testing for g ³ breakers.